PROJECT REFERENCE / CHAPTER 01

# HAWKER PROJECT DATA DICTIONARY

Seven-source MySQL database · Singapore hawker-centre opportunities
Reference version 1.0

Reference to the seven source datasets supporting a Singapore cooked-food stall tender and location-assessment project. The database links historical NEA tender appearances with centre locations, a community stall directory, URA planning context, 2020 census population, and MRT/LRT access. The table sections below document all physical MySQL fields.

| 11 | 112 | 12 | 2 |
|---|---|---|---|
| Physical tables | Physical fields | Foreign keys | SQL views |

## WHAT THE SEVEN SOURCES CONTRIBUTE

| SOURCE | DATASET | ROLE IN THE DATABASE |
|---|---|---|
| 1 | **NEA Hawker Centres GEOJSON** | Centre point, status, cooked-food capacity |
| 2 | **NEA Tender Notices** | Stall tender appearance, AMR, size, loads, fees |
| 3 | **Hawker Guide scrape** | Observed stalls, cuisines, directory occupancy |
| 4 | **MP2025 Land Use** | Zoned area/share in centre buffers |
| 5 | **MP2019 Subzone Boundaries (No Sea)** | Subzone reference; spatial centroid assignment |
| 6 | **Census Population 2020** | Subzone population by dwelling type |
| 7 | **LTA MRT Station Exits (includes LRT)** | MRT+LRT exit coordinates, nearest access |

### NOTES & DATA-QUALITY FLAGS

- Data are not a single-date snapshot: Census 2020, Master Plan 2019/2025, 2025–26 tenders and a recently scraped community directory describe different reference periods.
- This dictionary documents the reference SQL v1.0, reconstructed from the project DDL and confirmed rail_type addition. A live MySQL export remains authoritative for deployed definitions.

SECTION 00 / FIELD REFERENCE

# How to read this dictionary

A table-by-table field guide, modelled on the sample reference document.

Each table section gives its record grain, primary key, contributing source, and a four-column field reference. Example values illustrate expected formats; they are not guaranteed observations from the final database.

| HEADING | MEANING |
|---|---|
| **Column** | Exact MySQL column name. Join keys use the same spelling in SQL. |
| **Type** | Exact schema type plus NULL / NOT NULL; PK and FK are marked where applicable. |
| **Example** | Illustrative input or example category; NULL means a legitimate missing value. |
| **Description** | Meaning, units, provenance and rules for analysis. |

## KEY CONVENTIONS

- PK = primary key (unique row identifier). FK = foreign key (references another table). Composite PK means multiple columns jointly identify a row.
- NULL means unknown, unavailable, not recorded or not applicable; it is not the numeric value zero.
- A centre_id is the canonical key across sources. Government OBJECTID values and tender centre_key labels are external identifiers and need not be present for every centre.
- Shares such as land_use_share are fractional numbers (0 to approximately 1). Multiply by 100 only when presenting them as percentages.
- MRT-labelled table names are legacy names. Their current contents include both MRT and LRT; rail_type distinguishes them.

## DATABASE TABLES AT A GLANCE

| TABLE | GRAIN / WHAT ONE ROW MEANS | ROWS* |
|---|---|---|
| **source_registry** | 1 row per registered source | >= 7 |
| **subzones** | 1 row per MP2019 subzone | 332 |
| **hawker_centres** | 1 row per official NEA centre | 129 |
| **centre_aliases** | 1 row per source name and centre | >= 181 (indicative) |
| **stalls** | 1 row per (centre_id, normalized stall number) | 284 |
| **tender_opportunities** | 1 row per stall-month tender listing | 487 |
| **stall_observations** | 1 row per contributed observation | 4034 |
| **subzone_population** | 1 row per subzone and census year | 332 |
| **centre_land_use_metrics** | 1 row per centre, year, buffer, land-use category | 4079 |
| **mrt_exits** | 1 row per mapped exit | 613 |
| **centre_mrt_access** | 1 row per hawker centre | 129 |

* Expected / indicative counts from the project's validated imports and dictionary; use live SQL queries to confirm current counts.

SECTION 01 / FIELD REFERENCE

# source_registry

Source metadata

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per registered source | source_code |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| All seven | >= 7 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **source_code** | *VARCHAR(32)*<br>*NOT NULL*<br>**PK** | NEA_TENDER | Short controlled name for the contributing data source. |
| **source_title** | *VARCHAR(255)*<br>*NOT NULL* | NEA monthly tender notices | Human-readable name of data source. |
| **publisher** | *VARCHAR(120)*<br>*NOT NULL* | NEA | Government agency or directory provider responsible for source. |
| **source_url** | *VARCHAR(2048)*<br>*NULL* | nea.gov.sg/... | Original source webpage or public dataset URL. |
| **reference_period** | *VARCHAR(80)*<br>*NULL* | Oct 2025–Sep 2026 | Time period or plan vintage the dataset describes. |
| **source_notes** | *TEXT*<br>*NULL* | PDF extraction notes | Free text on provenance, cleaning conventions and caveats. |

### NOTES & DATA-QUALITY FLAGS

- This table records dataset provenance. One row per source code; rows are not stall observations.
- reference_period is informational and does not automatically make cross-source dates comparable.

SECTION 02 / FIELD REFERENCE

# subzones

URA MP2019 geographic reference

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per MP2019 subzone | subzone_id |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 5 | 332 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **subzone_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**PK** | 1 | Project-assigned Master Plan 2019 subzone identifier. |
| **planning_year** | *SMALLINT UNSIGNED*<br>*NOT NULL* | 2019 | Master Plan boundary vintage (2019). |
| **subzone_code** | *VARCHAR(30)*<br>*NOT NULL* | BMSZ12 | Subzone code as published by URA. |
| **subzone_name** | *VARCHAR(150)*<br>*NOT NULL* | DEPOT ROAD | URA 2019 subzone name. |
| **planning_area_code** | *VARCHAR(30)*<br>*NOT NULL* | BM | Planning-area code as published by URA. |
| **planning_area_name** | *VARCHAR(150)*<br>*NOT NULL* | BUKIT MERAH | URA 2019 planning-area name, used with subzone for Census reconciliation. |
| **region_name** | *VARCHAR(150)*<br>*NULL* | CENTRAL REGION | URA planning-region name. |

### NOTES & DATA-QUALITY FLAGS

- Subzones use Master Plan 2019 boundaries (no sea). URA polygon geometries stay in the offline GeoJSON; this SQL table stores codes and names.
- Planning-area code plus subzone code, within planning year, is a protected unique source reference.

SECTION 03 / FIELD REFERENCE

## hawker_centres

Canonical official hawker-centre master

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per official NEA centre | centre_id |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 1 (+ integration of 2, 3, 5) | 129 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **centre_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**PK** | 3 | Project-assigned stable centre identifier, shared across data sources. |
| **centre_key** | *VARCHAR(160)*<br>*NULL* | GOLDEN_MILE_FOOD_CENTRE | Optional project-level tender alias key. NULL is valid for centres with no unique tender alias; join using centre_id instead. |
| **nea_objectid** | *BIGINT*<br>*NULL* | 119479 | NEA GeoJSON OBJECTID used for source reconciliation; not the internal centre_id primary key. |
| **centre_name** | *VARCHAR(255)*<br>*NOT NULL* | Golden Mile Food Centre | Canonical display name of the hawker centre. |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **address** | *VARCHAR(500)*<br>*NULL* | 505 Beach Road | Source hawker-centre street address. |
| **postal_code** | *CHAR(6)*<br>*NULL* | 199583 | Singapore six-character postal code; store as text to preserve leading zeros. |
| **latitude** | *DECIMAL(10,7)*<br>*NULL* | 1.30... | Latitude in WGS84 decimal degrees (EPSG:4326). |
| **longitude** | *DECIMAL(10,7)*<br>*NULL* | 103.86... | Longitude in WGS84 decimal degrees (EPSG:4326). |
| **nea_status** | *VARCHAR(120)*<br>*NULL* | Existing | Latest administrative hawker-centre status in NEA source. |
| **number_of_cooked_food_stalls** | *SMALLINT UNSIGNED*<br>*NULL* | 100 | Published NEA number of cooked-food stalls at a centre. |
| **subzone_id** | *INT UNSIGNED*<br>*NULL*<br>**FK** | 1 | Project-assigned Master Plan 2019 subzone identifier. |
| **matched_to_nea** | *BOOLEAN*<br>*NOT NULL* | 1 (TRUE) | TRUE if centre was reconciled to a verified official NEA record. |

### NOTES & DATA-QUALITY FLAGS

- centre_id is the canonical join key. centre_key may be NULL when no unique tender-source label exists.
- latitude/longitude are WGS84 point coordinates; spatial distances were computed separately in projected metres.
- NEA OBJECTID is an official external ID; it is not the primary key of this database.

SECTION 04 / FIELD REFERENCE

## centre_aliases

Verified names in contributing sources

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per source name and centre | alias_id |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 1, 2, 3 | >= 181 (indicative) |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **alias_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**PK** | 1 | Internal surrogate identifier of a centre alias record. |
| **centre_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**FK** | 3 | Project-assigned stable centre identifier, shared across data sources. |
| **source_code** | *VARCHAR(32)*<br>*NOT NULL*<br>**FK** | NEA_TENDER | Short controlled name for the contributing data source. |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **centre_name_in_source** | *VARCHAR(255)*<br>*NOT NULL* | Golden Mile Food Centre | Centre name exactly as recorded in the relevant source (alias). |
| **matching_method** | *VARCHAR(50)*<br>*NULL* | reviewed_objectid | Evidence used for centre reconciliation, e.g. postal code or reviewed alias. |

### NOTES & DATA-QUALITY FLAGS

- Multiple names from different sources can legitimately map to one official centre_id.
- Review and verify aliases before import; avoid using fuzzy name matching at query time.

SECTION 05 / FIELD REFERENCE

## stalls

Physical stall dimension for tender history

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per (centre_id, normalized stall number) | stall_id |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 2 | 284 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **stall_id** | *BIGINT UNSIGNED*<br>*NOT NULL*<br>**PK** | 1 | Project-assigned identifier of a physical stall (centre and unit number). |
| **centre_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**FK** | 3 | Project-assigned stable centre identifier, shared across data sources. |
| **stall_no_normalised** | *VARCHAR(40)*<br>*NOT NULL* | #01-001 | Normalized stall/unit number including floor and any letter suffix; unique only within a centre. |

### NOTES & DATA-QUALITY FLAGS

- One physical tender stall can appear in multiple monthly tender_opportunities rows.
- Hawker Guide observations are not guaranteed to match a tender stall_id; a NULL reference can be valid.

SECTION 06 / FIELD REFERENCE

## tender_opportunities

NEA monthly cooked-food stall listings

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per stall-month tender listing | listing_id |

| SOURCE(S) | EXPECTED RECORDS |
|---|---|
| 2 | 487 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **listing_id** | *BIGINT UNSIGNED*<br>*NOT NULL*<br>**PK** | 1 | Unique tender appearance identifier from validated tender CSV. |
| **stall_id** | *BIGINT UNSIGNED*<br>*NOT NULL*<br>**FK** | 1 | Project-assigned identifier of a physical stall (centre and unit number). |
| **centre_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**FK** | 3 | Project-assigned stable centre identifier, shared across data sources. |
| **exercise_month** | *DATE*<br>*NOT NULL* | 2025-10-01 | First calendar day of NEA monthly tender exercise (CSV YYYY-MM normalized to YYYY-MM-01). |
| **tender_open_date** | *DATE*<br>*NULL* | 2025-10-13 | Official date when the NEA monthly tender exercise opened. |
| **tender_close_date** | *DATE*<br>*NULL* | 2025-10-27 | Official date when the NEA monthly tender exercise closed. |
| **stall_no** | *VARCHAR(40)*<br>*NULL* | #02-80 | Stall number as printed in tender notice. |
| **release_category** | *VARCHAR(40)*<br>*NULL* | second_or_subsequent | Published NEA category: previously released no bids, first release, or subsequent release. |
| **trade_marker** | *VARCHAR(30)*<br>*NULL* | + / ^ | Original symbols following stall number, for example + or ^. |
| **allowed_trade_hint** | *VARCHAR(255)*<br>*NULL* | subject_to_pdf_notes | Broad interpretation of allowed trade; inspect official notes for definitive restrictions. |
| **approximate_area_m2** | *DECIMAL(9,2)*<br>*NULL* | 5.65 | Approximate stall floor area in square metres from monthly PDF. |
| **area_verified** | *BOOLEAN*<br>*NOT NULL* | 1 (TRUE) | Whether the extracted stall area could be independently assigned in the PDF. |
| **electrical_load_amps** | *DECIMAL(8,2)*<br>*NULL* | 17.00 | Approved electrical design load in amperes, if extracted and verified. |
| **electrical_verified** | *BOOLEAN*<br>*NOT NULL* | 1 (TRUE) | Whether the electrical load was verified at the correct stall in PDF. |
| **amr_reference_year** | *SMALLINT UNSIGNED*<br>*NULL* | 2025 | Reference year for the displayed latest median assessed market rent. |
| **median_centre_amr_sgd** | *DECIMAL(12,2)*<br>*NULL* | 1,450.00 | NEA centre-level median assessed market rent reference (SGD), not actual stall rent, winning bid, or payable total. |
| **amr_2023_sgd** | *DECIMAL(12,2)*<br>*NULL* | 1,450.00 | Published centre-level median AMR for 2023 (SGD before GST), if available. |
| **amr_2024_sgd** | *DECIMAL(12,2)*<br>*NULL* | 1,450.00 | Published centre-level median AMR for 2024 (SGD before GST), if available. |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **amr_2025_sgd** | *DECIMAL(12,2) NULL* | 1,450.00 | Published centre-level median AMR for 2025 (SGD before GST), if available. |
| **amr_2026_sgd** | *DECIMAL(12,2) NULL* | NULL | Published centre-level median AMR for 2026 (SGD before GST), if available. |
| **cleaning_fee_raw** | *VARCHAR(200) NULL* | $400 to $500 | Exact cleaning-fee wording/amount extracted from monthly tender PDF. |
| **cleaning_fee_kind** | *VARCHAR(40) NULL* | range | Fee representation: range, greater-than limit, or special case. |
| **cleaning_fee_min_sgd** | *DECIMAL(12,2) NULL* | 400.00 | Lower endpoint when cleaning fee is a bounded range; not a quoted exact fee. |
| **cleaning_fee_max_sgd** | *DECIMAL(12,2) NULL* | 500.00 | Upper endpoint when cleaning fee is a bounded range; not a quoted exact fee. |
| **cleaning_fee_strict_lower_bound_sgd** | *DECIMAL(12,2) NULL* | NULL | Threshold when PDF states cleaning fee greater than a stated amount. |
| **scc_raw** | *VARCHAR(200) NULL* | $150 to $200 | Exact service and conservancy charge wording/amount from monthly tender PDF. |
| **scc_kind** | *VARCHAR(40) NULL* | range | Charge representation: range, greater-than limit, or special case. |
| **scc_min_sgd** | *DECIMAL(12,2) NULL* | 150.00 | Lower endpoint of published service and conservancy charge range. |
| **scc_max_sgd** | *DECIMAL(12,2) NULL* | 200.00 | Upper endpoint of published service and conservancy charge range. |
| **scc_strict_lower_bound_sgd** | *DECIMAL(12,2) NULL* | NULL | Threshold when published S&CC exceeds a stated amount. |
| **costs_need_review** | *BOOLEAN NOT NULL* | 0 (FALSE) | TRUE if fee or S&CC could not be interpreted reliably. |
| **needs_review** | *BOOLEAN NOT NULL* | 0 (FALSE) | TRUE if any field in the tender listing requires manual review. |
| **raw_stall_token** | *VARCHAR(120) NULL* | #02-80 | Original stall-number text including symbols from source PDF. |
| **stall_note_context** | *TEXT NULL* | refer to PDF notes | Published note references associated with the stall; do not assume restrictions from cuisine alone. |
| **source_file** | *VARCHAR(255) NOT NULL* | oct-2025-tender-notice.pdf | Exact monthly source PDF filename for tracing tender record. |
| **source_page** | *SMALLINT UNSIGNED NULL* | 2 | 1-indexed PDF page number where tender listing appears. |

### NOTES & DATA-QUALITY FLAGS

- One row represents a cooked-food stall appearance in one NEA tender exercise, not a successful letting or a currently available stall.
- median_centre_amr_sgd is centre-level assessed market rent, not an awarded bid or exact stall rent. Treat fee ranges and strict lower bounds as interval/threshold data.

- Some areas/electrical loads are NULL because the PDF layout prevented independent verification. Flags identify records requiring review.
- Trade markers and restrictions should be read with their source PDF notes; do not infer licence eligibility from cuisine alone.

SECTION 07 / FIELD REFERENCE

# stall_observations

Hawker Guide directory snapshot

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per contributed observation | observation_id |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 3 | 4034 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **observation_id** | *BIGINT UNSIGNED NOT NULL* **PK** | 1 | Unique surrogate key for each scraped stall observation. |
| **centre_id** | *INT UNSIGNED NOT NULL* **FK** | 3 | Project-assigned stable centre identifier, shared across data sources. |
| **stall_id** | *BIGINT UNSIGNED NULL* **FK** | NULL | Project-assigned identifier of a physical stall (centre and unit number). |
| **observed_on** | *DATE NULL* | NULL | Actual scrape/observation date, if independently recorded; SQL NULL for the current undated community scrape. |
| **unit_no_raw** | *VARCHAR(40) NULL* | #01-01 | Original Hawker Guide stall/unit number, including formatting. |
| **unit_no_normalised** | *VARCHAR(40) NULL* | #01-001 / NULL | Normalized Hawker Guide unit number. NULL also marks deliberately quarantined conflicting unit records; do not count these as verified distinct units. |
| **occupancy_status** | *VARCHAR(40) NULL* | occupied | Community directory-reported occupied/vacant state; not official NEA occupancy or tender availability. |
| **store_name** | *VARCHAR(255) NULL* | Stall display name | Directory name of operating food stall, when observed. |
| **cuisine_raw** | *VARCHAR(150) NULL* | Chinese | Original scraped cuisine label. |
| **cuisine_category** | *VARCHAR(150) NULL* | Chinese | Standardized cuisine label used to count local competition. |
| **price_range_raw** | *VARCHAR(100) NULL* | $4–$8 | Directory price range as reported; may not represent actual transaction prices. |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **postal_code_1** | *CHAR(6)*<br>*NULL* | 048940 | First postal code as contributed by Hawker Guide; not an authoritative location key. Use hawker_centres.postal_code for official values. |
| **postal_code_2** | *CHAR(6)*<br>*NULL* | NULL | Second postal code as contributed by Hawker Guide, if present; may disagree with the official NEA record. |
| **source_url** | *VARCHAR(2048)*<br>*NOT NULL* | hawker.guide/... | Original source webpage or public dataset URL. |

### NOTES & DATA-QUALITY FLAGS

- Community-contributed directory snapshot. observed_on is NULL for the current scrape because its exact date was not recorded.
- A NULL unit_no_normalised may reflect deliberately quarantined conflicting units; such rows are excluded from the cuisine aggregation view.
- Directory occupancy and cuisine are observations, not official NEA vacancy declarations; source postal codes are not authoritative location keys.

SECTION 08 / FIELD REFERENCE

## subzone_population

2020 Census demographics

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per subzone and census year | (subzone_id, census_year) |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 6 | 332 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **subzone_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**PK/FK** | 1 | Project-assigned Master Plan 2019 subzone identifier. |
| **census_year** | *SMALLINT UNSIGNED*<br>*NOT NULL*<br>**PK component** | 2020 | Census reference year (2020 for project). |
| **resident_population** | *INT UNSIGNED*<br>*NULL* | 12,345 | Number of residents in selected planning subzone from Census 2020. |
| **hdb_population** | *INT UNSIGNED*<br>*NULL* | 9,120 | Residents in HDB dwellings, all room types combined. |
| **hdb_1_2_room** | *INT UNSIGNED*<br>*NULL* | 450 | Residents of HDB 1-room and 2-room flats. |
| **hdb_3_room** | *INT UNSIGNED*<br>*NULL* | 1,200 | Residents of HDB 3-room flats. |
| **hdb_4_room** | *INT UNSIGNED*<br>*NULL* | 4,000 | Residents of HDB 4-room flats. |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **hdb_5_room_executive** | *INT UNSIGNED NULL* | 3,470 | Residents of HDB 5-room and executive flats. |
| **condo_other_apartments** | *INT UNSIGNED NULL* | 1,200 | Residents in condominiums and other apartments. |
| **landed_properties** | *INT UNSIGNED NULL* | 800 | Residents in landed properties. |
| **other_dwellings** | *INT UNSIGNED NULL* | 1,225 | Residents in other types of dwelling. |

### NOTES & DATA-QUALITY FLAGS

- 2020 Census counts attach to the Master Plan 2019 subzone geography by code. They are not current residents or estimates of visitors within walking distance.
- Suppressed, unavailable or unreported counts remain NULL; never replace those values with zero.

SECTION 09 / FIELD REFERENCE

## centre_land_use_metrics

2025 zoning around centres

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per centre, year, buffer, land-use category | (centre_id, plan_year, buffer_radius_m, land_use_category) |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 4 | 4079 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **centre_id** | *INT UNSIGNED NOT NULL* **PK/FK** | 3 | Project-assigned stable centre identifier, shared across data sources. |
| **plan_year** | *SMALLINT UNSIGNED NOT NULL* **PK component** | 2025 | Land-use plan vintage (2025 for project). |
| **buffer_radius_m** | *SMALLINT UNSIGNED NOT NULL* **PK component** | 500 / 1,000 | Radius in metres used for geographical buffer around a hawker centre. |
| **land_use_category** | *VARCHAR(100) NOT NULL* **PK component** | RESIDENTIAL | Normalized zoning class such as RESIDENTIAL or COMMERCIAL. |
| **zoned_area_m2** | *DECIMAL(16,2) NULL* | 135,600.00 | Area of zoning polygons intersecting selected centre buffer (square metres). |
| **land_use_share** | *DECIMAL(9,6) NULL* | 0.172650 | Land-use category polygon area / full circular buffer area (fraction, not percent). Zoning coverage may be below 100%, particularly near the sea; not footfall. |

### NOTES & DATA-QUALITY FLAGS

- Buffer radius is 500 m or 1,000 m around each centre point. Each category is a separate row, so this is not one row per centre.
- land_use_share = zoned area in that category / full circular buffer area. Coastal and excluded areas can leave the sum below 1.
- Land use is planning-zone designation, not measured resident density or retail footfall.

SECTION 10 / FIELD REFERENCE

# mrt_exits

LTA MRT + LRT exits

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per mapped exit | exit_id |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 7 | 613 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **exit_id** | *INT UNSIGNED*<br>*NOT NULL*<br>**PK** | 1 | Internal unique ID of an LTA MRT or LRT station exit; referenced by centre_mrt_access.nearest_exit_id. |
| **lta_objectid** | *BIGINT*<br>*NULL* | 21404 | External LTA OBJECTID for an MRT or LRT exit; not the internal primary key. |
| **station_name** | *VARCHAR(255)*<br>*NOT NULL* | SPRINGLEAF MRT STATION | LTA station label for an MRT or LRT exit; some labels are station codes (e.g., CC30). |
| **exit_code** | *VARCHAR(100)*<br>*NULL* | Exit 1 | Published exit label, such as Exit 1, Exit A, etc. |
| **rail_type** | *VARCHAR(3)*<br>*NULL* | MRT / LRT | MRT or LRT, classified from LTA station names/codes. Distinguishes transport modes for visual filters; this does not imply relative passenger traffic. |
| **latitude** | *DECIMAL(10,7)*<br>*NOT NULL* | 1.3987872 | Latitude in WGS84 decimal degrees (EPSG:4326). |
| **longitude** | *DECIMAL(10,7)*<br>*NOT NULL* | 103.8180034 | Longitude in WGS84 decimal degrees (EPSG:4326). |

### NOTES & DATA-QUALITY FLAGS

- This table contains 613 MRT and LRT station exits in the fixed study dataset (541 MRT, 72 LRT). rail_type enables mapping filters.
- Some station_name values are LTA codes (e.g., CC30) rather than full human-friendly station names.
- An exit is not a unique station. Count distinct station names when summarising station access.

SECTION 11 / FIELD REFERENCE

# centre_mrt_access

Centre-to-rail proximity summaries

| RECORD GRAIN | PRIMARY KEY |
|---|---|
| 1 row per hawker centre | centre_id |
| **SOURCE(S)** | **EXPECTED RECORDS** |
| 1+7 | 129 |

| COLUMN | MYSQL TYPE | EXAMPLE | DESCRIPTION |
|---|---|---|---|
| **centre_id** | *INT UNSIGNED NOT NULL* **PK/FK** | 3 | Project-assigned stable centre identifier, shared across data sources. |
| **nearest_exit_id** | *INT UNSIGNED NULL* **FK** | 1 | Nearest MRT or LRT exit after combining both transport types; FK to mrt_exits.exit_id. |
| **nearest_exit_distance_m** | *DECIMAL(12,2) NULL* | 320.50 | Straight-line distance in metres from centre coordinate to nearest MRT or LRT exit; not walking route distance. |
| **exits_within_500m** | *INT UNSIGNED NULL* | 4 | Count of all MRT and LRT exits within a straight-line radius of 500 metres. |
| **stations_within_500m** | *INT UNSIGNED NULL* | 2 | Distinct MRT and LRT station names having an exit within 500 m; multiple exits for one station count once. |

## NOTES & DATA-QUALITY FLAGS

- Nearest exit and 500 m counts combine MRT and LRT. They are Euclidean (straight-line) spatial measurements, not route-based travel time.
- A filter to MRT-only or LRT-only cannot be applied directly to this combined summary; recompute from mrt_exits points for mode-specific distance.

SECTION 12 / FIELD REFERENCE

# How the tables fit together

Canonical centre IDs, tender stall identities and geographical reference keys.

The physical schema contains 12 declared foreign keys. Use these relationships to enrich tender listings without inventing joins based on similar names.

| CHILD.TABLE → COLUMN | PARENT.TABLE → COLUMN | DELETE RULE |
|---|---|---|
| **hawker_centres.subzone_id** | subzones.subzone_id | SET NULL |
| **centre_aliases.centre_id** | hawker_centres.centre_id | RESTRICT |
| **centre_aliases.source_code** | source_registry.source_code | RESTRICT |
| **stalls.centre_id** | hawker_centres.centre_id | RESTRICT |
| **tender_opportunities.stall_id, centre_id** | stalls.stall_id, centre_id | RESTRICT |
| **tender_opportunities.centre_id** | hawker_centres.centre_id | RESTRICT |
| **stall_observations.centre_id** | hawker_centres.centre_id | RESTRICT |
| **stall_observations.stall_id, centre_id** | stalls.stall_id, centre_id | RESTRICT |
| **subzone_population.subzone_id** | subzones.subzone_id | RESTRICT |
| **centre_land_use_metrics.centre_id** | hawker_centres.centre_id | RESTRICT |
| **centre_mrt_access.centre_id** | hawker_centres.centre_id | CASCADE |
| **centre_mrt_access.nearest_exit_id** | mrt_exits.exit_id | SET NULL |

### NOTES & DATA-QUALITY FLAGS

- hawker_centres.centre_id is the central integration key. Centre aliases reconcile different source names to this primary key.
- tender_opportunities is at listing-level grain (487 rows); joining it to a multi-row land-use or cuisine table without first aggregating will multiply rows.
- subzone_population is keyed by subzone_id plus census_year; the census year is fixed at 2020 for this project.

# TWO ANALYTICAL SQL VIEWS

## v_tender_dashboard

One row per tender listing; left-joins centre, 2019 subzone, 2020 population, combined MRT/LRT access and 2025 land-use shares

| GRAIN | EXPECTED ROWS | CAUTION |
|---|---|---|
| listing_id | 487 | Do not SUM centre/subzone data over listing rows without deduplicating centres. `nearest_mrt_station` is a legacy alias that can point to LRT. |

## v_cuisine_mix

Counts latest classified occupied community stall observations per centre and cuisine

| GRAIN | EXPECTED ROWS | CAUTION |
|---|---|---|
| centre_id + cuisine_category | Variable | Only normalized non-NULL unit numbers included (quarantine excluded). Source is undated; not 2025/26 historical. |

# COMMON QUERY PATHS

| ANALYSIS | PREFERRED SOURCE | JOIN / GRAIN RULE |
|---|---|---|
| **Compare tender opportunities** | v_tender_dashboard | One row per listing_id |
| **Compare cuisine competition** | v_cuisine_mix | One row per centre_id and cuisine_category |
| **Explore subzone demographics** | hawker_centres + subzones + subzone_population | Join on subzone_id; restrict census_year to 2020 |
| **Compare MRT-only vs LRT-only** | mrt_exits + hawker_centres | Filter rail_type; calculate distances separately |

# EXAMPLE: TENDER-LEVEL SQL

```
SELECT listing_id, centre_name, median_centre_amr_sgd
FROM v_tender_dashboard
WHERE median_centre_amr_sgd IS NOT NULL
LIMIT 10;
```

### NOTES & DATA-QUALITY FLAGS

- For centre-level statistics, aggregate or deduplicate the tender-level view by centre_id before summing repeated centre/subzone attributes.

SECTION 13 / FIELD REFERENCE

# Interpretation, units & data quality

A reference for reporting valid comparisons in SQL, R and final visualisations.

## 01 Time is not uniform

- Tenders: October 2025–September 2026, one row per listed stall in a monthly exercise.
- Census population: 2020. URA subzone boundaries: Master Plan 2019. URA zoning: Master Plan 2025.
- Hawker Guide: recently scraped community-contributed snapshot; exact observed_on is unknown.

## 02 Cost semantics

- Median assessed market rent (AMR) is a centre-level rental reference, not the awarded bid, current stall rent or total monthly operating cost.
- Cleaning charges and service & conservancy charges may be bounded ranges, strictly greater-than amounts, or special cases. Do not silently replace ranges with a single exact charge.
- Source values may be before GST. Do not combine monetary fields without reviewing comparability and reference period.

## 03 Spatial metrics

- Distances to station exits are straight-line metres measured in EPSG:3414, not walking distances or travel times.
- An MRT/LRT exit is not a station. Use distinct-station counts when appropriate, not just number of exit points.
- Land-use shares divide polygon area by the full circular buffer, not by mapped land area. A centre near the coast can have total coverage well below 100%.

## 04 Missing, unverified and quarantined

- NULL is not zero. Use SQL IS NULL or IS NOT NULL; unknown areas, loads, dates and suppressed counts should remain missing.
- area_verified, electrical_verified, costs_need_review and needs_review record uncertainty in tender extraction.
- Quarantined Hawker Guide units retain raw strings while unit_no_normalised is NULL. Do not assume unlisted vendors are absent or vacant.

## 05 Usefulness and joins

- Start from one record grain: centre, stall, tender listing, subzone, or centre/category/radius. Aggregate first before joining across grains.
- Use v_tender_dashboard for tender-level analysis; use v_cuisine_mix for classified observed cuisine mix, not historical vacancy.
- For MRT-only/LRT-only accessibility, calculate type-specific proximity from the exits table rather than reusing combined summary counts.

SECTION 14 / FIELD REFERENCE

# Source catalogue & handoff

File provenance and the role of each offline study dataset.

## 1. NEA Hawker Centres GEOJSON

**Project file:** HawkerCentresGEOJSON.geojson
**Destination:** hawker_centres
**Scope:** Official master data; own reference date

https://data.gov.sg/datasets/d_4a086da0a5553be1d89383cd90d07ecd/view

## 2. NEA Tender Notices

**Project file:** tender_extracted_oct25tosep26.csv
**Destination:** stalls; tender_opportunities; centre_aliases
**Scope:** Oct 2025–Sep 2026, AMR ≠ actual rent

https://www.nea.gov.sg/corporate-functions/resources/tender-notices

## 3. Hawker Guide scrape

**Project file:** hawker_guide_all_stalls.csv
**Destination:** stall_observations; centre_aliases
**Scope:** Undated snapshot; missing code quarantine

https://hawker.guide/

## 4. MP2025 Land Use

**Project file:** MasterPlan2025LandUseLayer.geojson
**Destination:** centre_land_use_metrics
**Scope:** 2025 zoning, not observed footfall

https://data.gov.sg/datasets/d_a8c3546b26712e35021f3a681d0353ae/view

## 5. MP2019 Subzone Boundaries (No Sea)

**Project file:** MasterPlan2019SubzoneBoundaryNoSeaGEOJSON.geojson
**Destination:** subzones; hawker_centres.subzone_id
**Scope:** 2019 boundaries, no sea area

https://data.gov.sg/datasets/d_8594ae9ff96d0c708bc2af633048edfb/view

## 6. Census Population 2020

**Project file:** ResidentPopulationbyPlanningAreaSubzoneofResidenceandTypeofDwellingCensusofPopulation2020.csv
**Destination:** subzone_population
**Scope:** 2020 residents, not modern catchment count

https://data.gov.sg/datasets/d_7f243956483d5901f237e6f87b096636/view

## 7. LTA MRT Station Exits (includes LRT)

**Project file:** LTAMRTStationExitGEOJSON.geojson
**Destination:** mrt_exits; centre_mrt_access
**Scope:** Straight-line distances, not pedestrian time

https://data.gov.sg/datasets/d_b39d3a0871985372d7e1637193335da5/view